

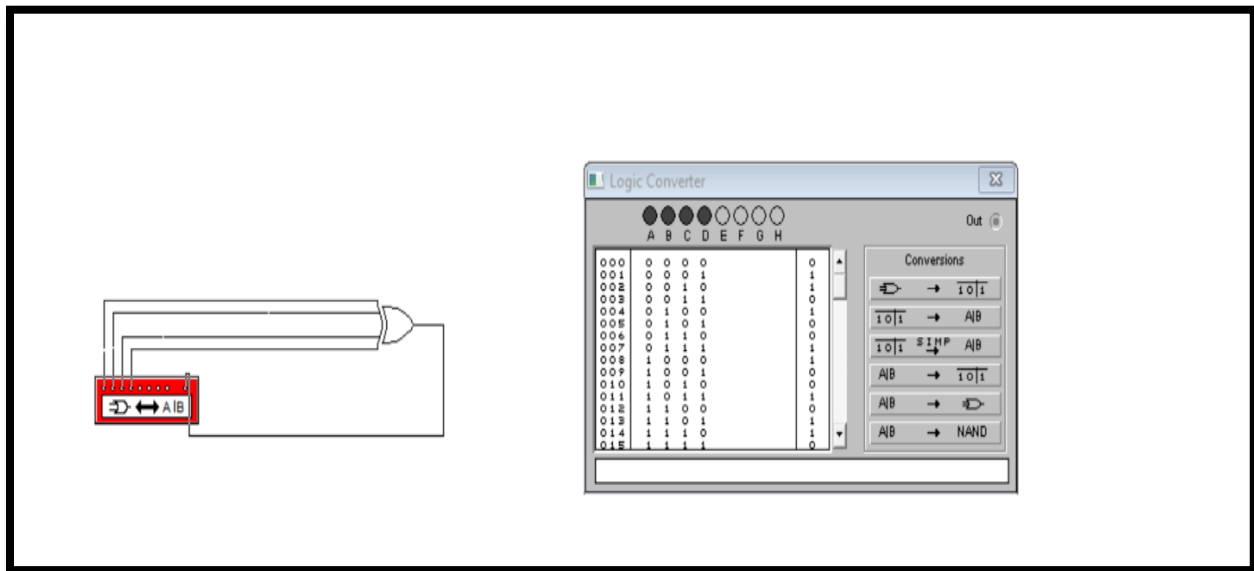


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Semester		2nd
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Subject		CA&LD
Course Code		CS-1206
Credit Hours		3+1
Lab:		Open Ended Lab
Date		22-Sep-2025
Department of Software Engineering		

Task 1: Parity Generator & Checker (PLA / Logic Simulator)

1. Parity Generator Circuit

- **Description:** 4-bit even parity generator ($A, B, C, D \rightarrow P$).
- **Screenshot 1: Parity Generator Circuit**



2. Truth Table Verification (Generator)

3. Parity Checker Circuit

- **Description:** 4-bit data word + parity $\rightarrow$ Error output.
- **Screenshot 3: Parity Checker Circuit**

4. Checker (No Error Case)

Checker output:

$$\text{ERROR} = A \oplus B \oplus C \oplus D \oplus P \text{ ERROR} = A \oplus B \oplus C \oplus D \oplus P$$

$$\text{PERROR} = A \oplus B \oplus C \oplus D \oplus P$$

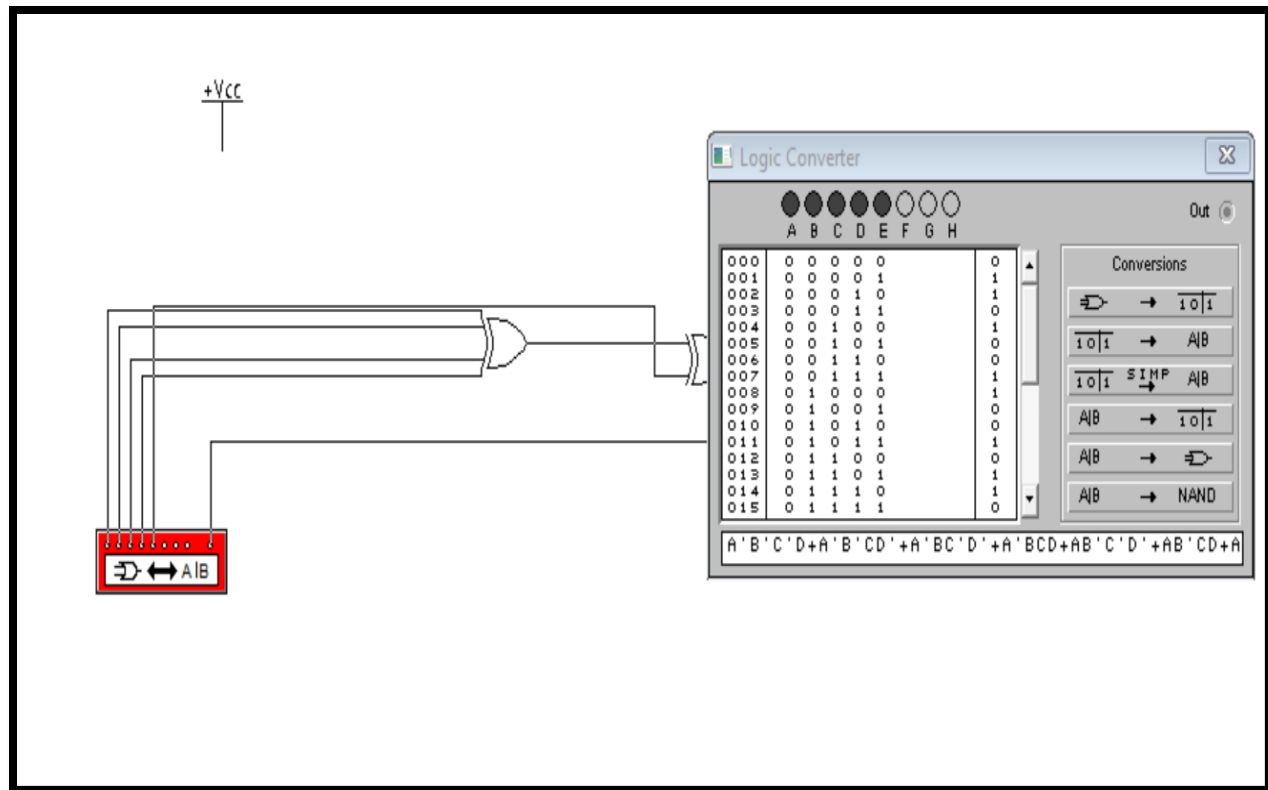
- If total number of 1s is **even** → **ERROR = 0**
 - If total number of 1s is **odd** → **ERROR = 1**
-

◆ Checker Table (rows 1–10)

Row	A	B	C	D	P	Total 1's	ERROR
1	0	0	0	0	0	0 → even	0
2	0	0	0	0	1	1 → odd	1
3	0	0	0	1	0	1 → odd	1
4	0	0	0	1	1	2 → even	0
5	0	0	1	0	0	1 → odd	1
6	0	0	1	0	1	2 → even	0
7	0	0	1	1	0	2 → even	0
8	0	0	1	1	1	3 → odd	1
9	0	1	0	0	0	1 → odd	1
10	0	1	0	0	1	2 → even	0

- **Description:** With correct parity bit → E=0.

Screenshot 4: Checker – No Error



5. Checker (Error Case)

- **Description:** Flip one input (A, B, C, D, or P) $\rightarrow E=1$.
- **Screenshot 5: Checker – Error Detected**

6. Integrated Test (Optional)

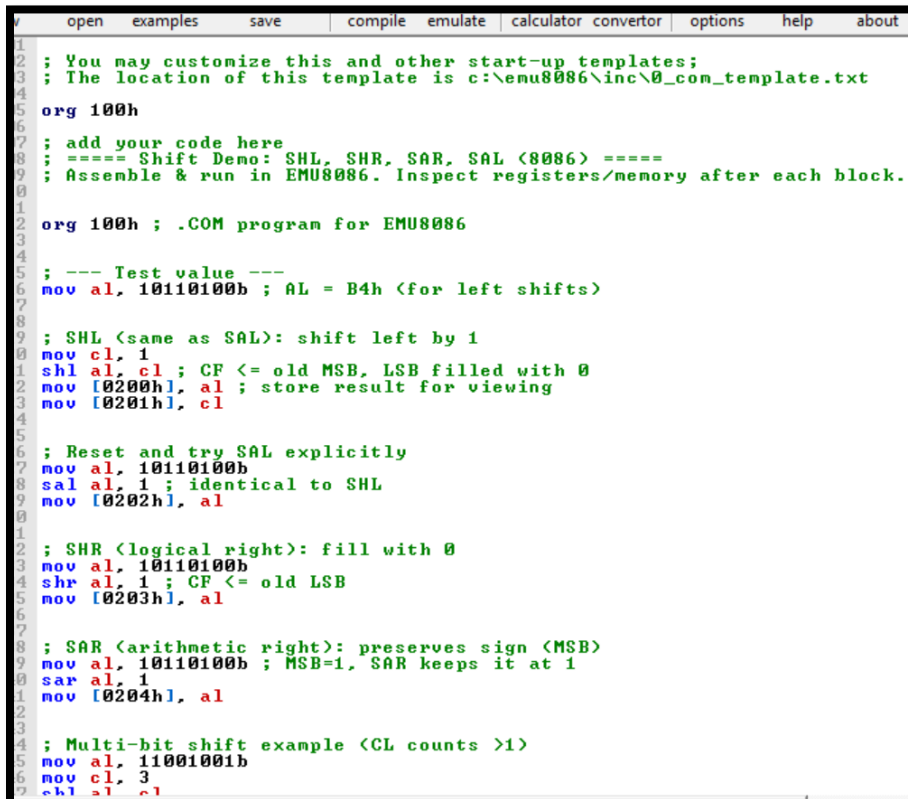
- **Description:** Full system: generator + checker combined.
- **Screenshot 6: Integrated Generator + Checker Test**

◆ Task 2: Shift Instructions (EMU8086)

1. Program Listing

- **Description:** Assembly code written in EMU8086.

Screenshot 7: Code in EMU8086



```
1  ; You may customize this and other start-up templates;
2  ; The location of this template is c:\emu8086\inc\0_com_template.txt
3
4
5  org 100h
6
7  ; add your code here
8  ; ===== Shift Demo: SHL, SHR, SAR, SAL (8086) =====
9  ; Assemble & run in EMU8086. Inspect registers/memory after each block.
10
11
12  org 100h ; .COM program for EMU8086
13
14  ; --- Test value ---
15  mov al, 10110100b ; AL = B4h (for left shifts)
16
17
18  ; SHL (same as SAL): shift left by 1
19  mov cl, 1
20  shl al, cl ; CF <= old MSB, LSB filled with 0
21  mov [0200h], al ; store result for viewing
22  mov [0201h], cl
23
24
25  ; Reset and try SAL explicitly
26  mov al, 10110100b
27  sal al, 1 ; identical to SHL
28  mov [0202h], al
29
30
31  ; SHR (logical right): fill with 0
32  mov al, 10110100b
33  shr al, 1 ; CF <= old LSB
34  mov [0203h], al
35
36
37  ; SAR (arithmetic right): preserves sign (MSB)
38  mov al, 10110100b ; MSB=1, SAR keeps it at 1
39  sar al, 1
40  mov [0204h], al
41
42
43  ; Multi-bit shift example (CL counts >1)
44  mov al, 11001001b
45  mov cl, 3
46  shl al, cl
```

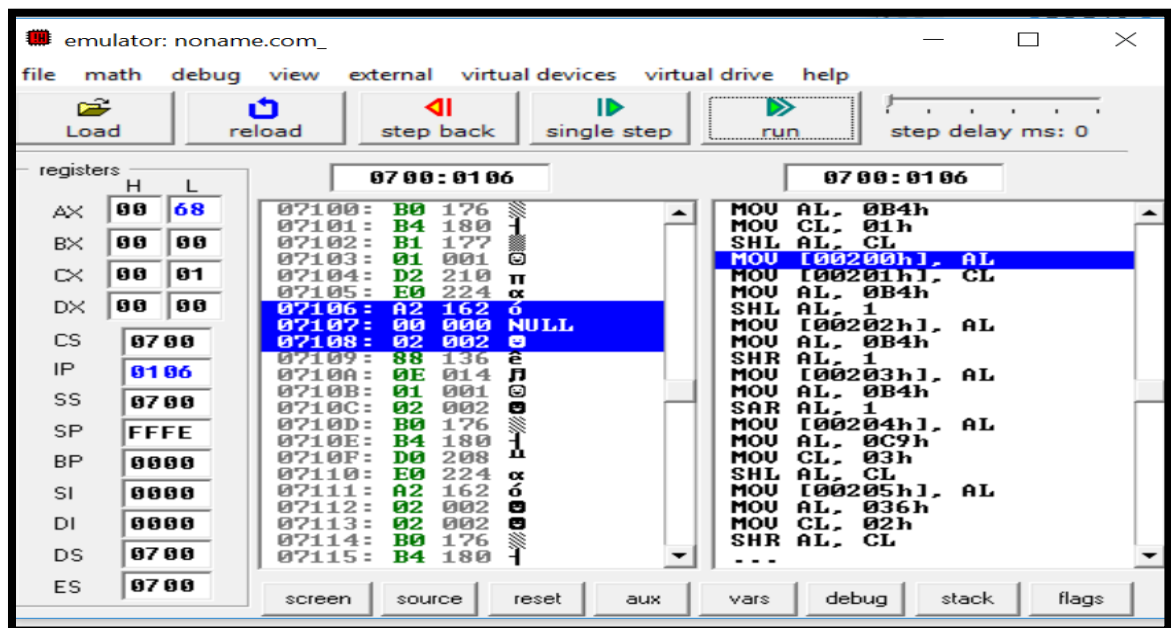
```

22 mov [0200h], al ; store result for viewing
23 mov [0201h], cl
24
25 ; Reset and try SAL explicitly
26
27 mov al, 10110100b
28 sal al, 1 ; identical to SHL
29 mov [0202h], al
30
31 ; SHR (logical right): fill with 0
32
33 mov al, 10110100b
34 shr al, 1 ; CF <= old LSB
35 mov [0203h], al
36
37 ; SAR (arithmetic right): preserves sign (MSB)
38
39 mov al, 10110100b ; MSB=1, SAR keeps it at 1
40 sar al, 1
41 mov [0204h], al
42
43 ; Multi-bit shift example (CL counts >1)
44
45 mov al, 11001001b
46 mov cl, 3
47 shl al, cl
48 mov [0205h], al
49
50 ; Another right shift with CL
51
52 mov al, 00110110b
53 mov cl, 2
54 shr al, cl
55 mov [0206h], al
56
57 ; Exit to DOS
58 mov ah, 4Ch
59 int 21h
60
61 ret
62
63
64
65
66
67

```

2. SHL AL,1

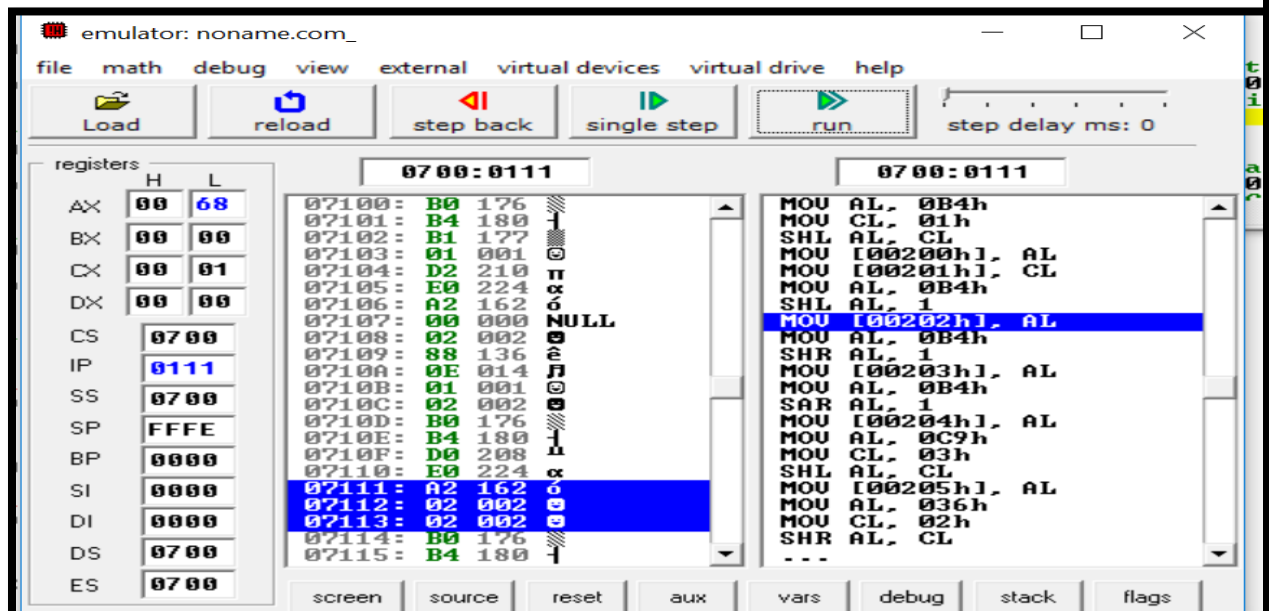
- Expected Output: AL=68h, CF=1, OF=1.
- Screenshot 8: Registers/Flags after SHL



3. SAL AL,1

- Expected Output: AL=68h (same as SHL).

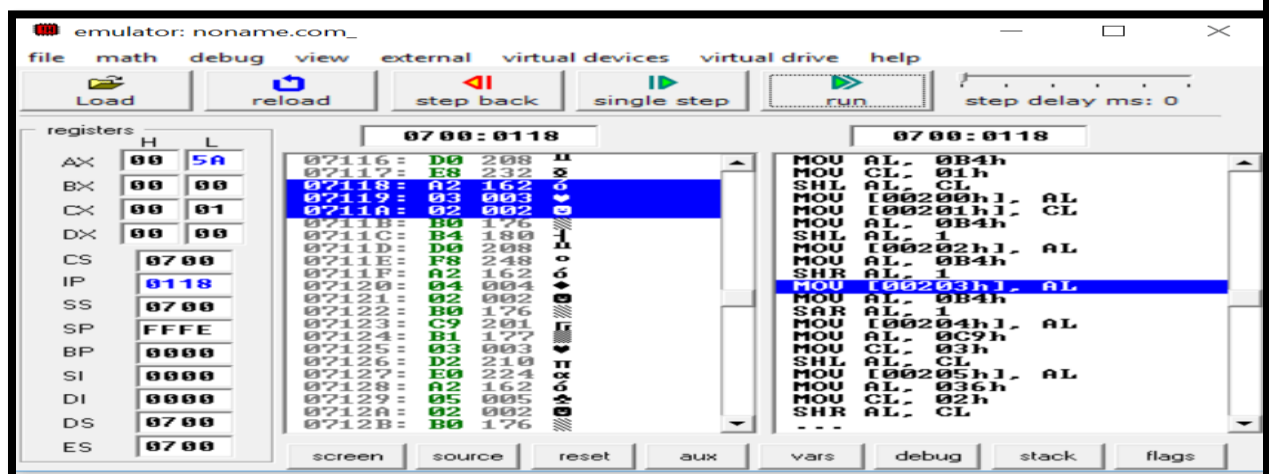
Screenshot 9: Registers/Flags after SAL



4. SHR AL,1

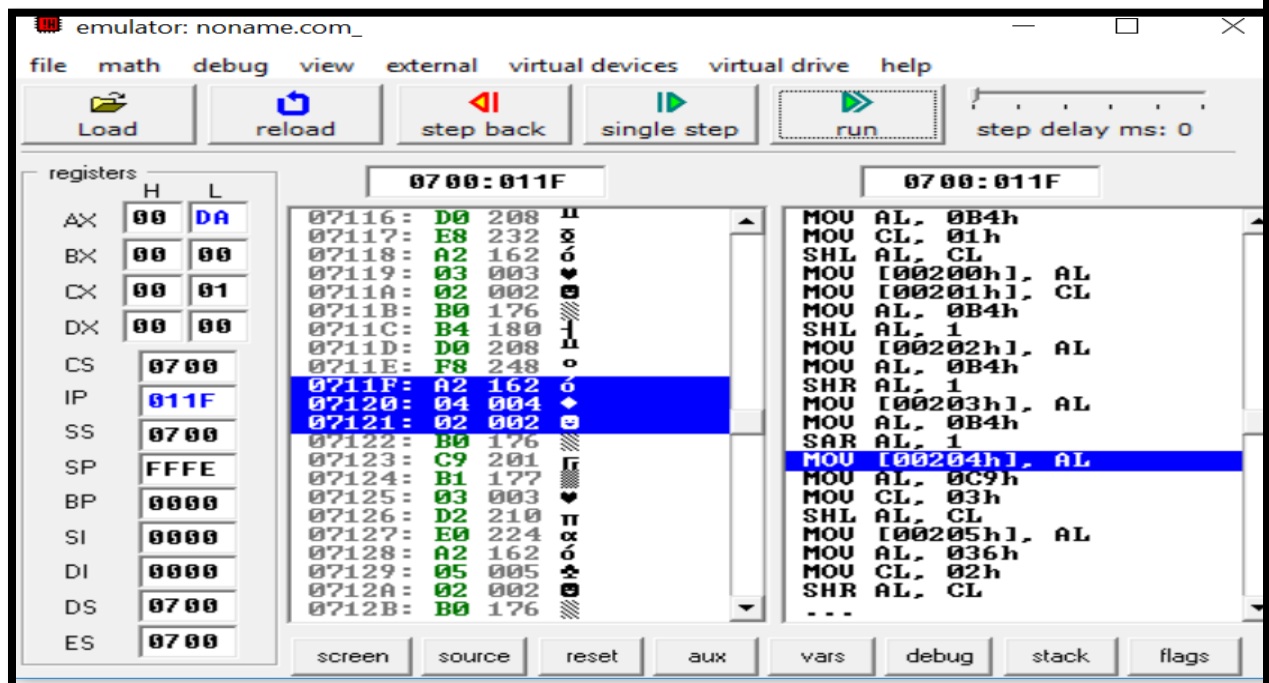
- Expected Output: AL=5Ah, CF=0, OF=1.
- (Insert screenshot of Registers + Flags after SHR)

Screenshot 10: Registers/Flags after SHR



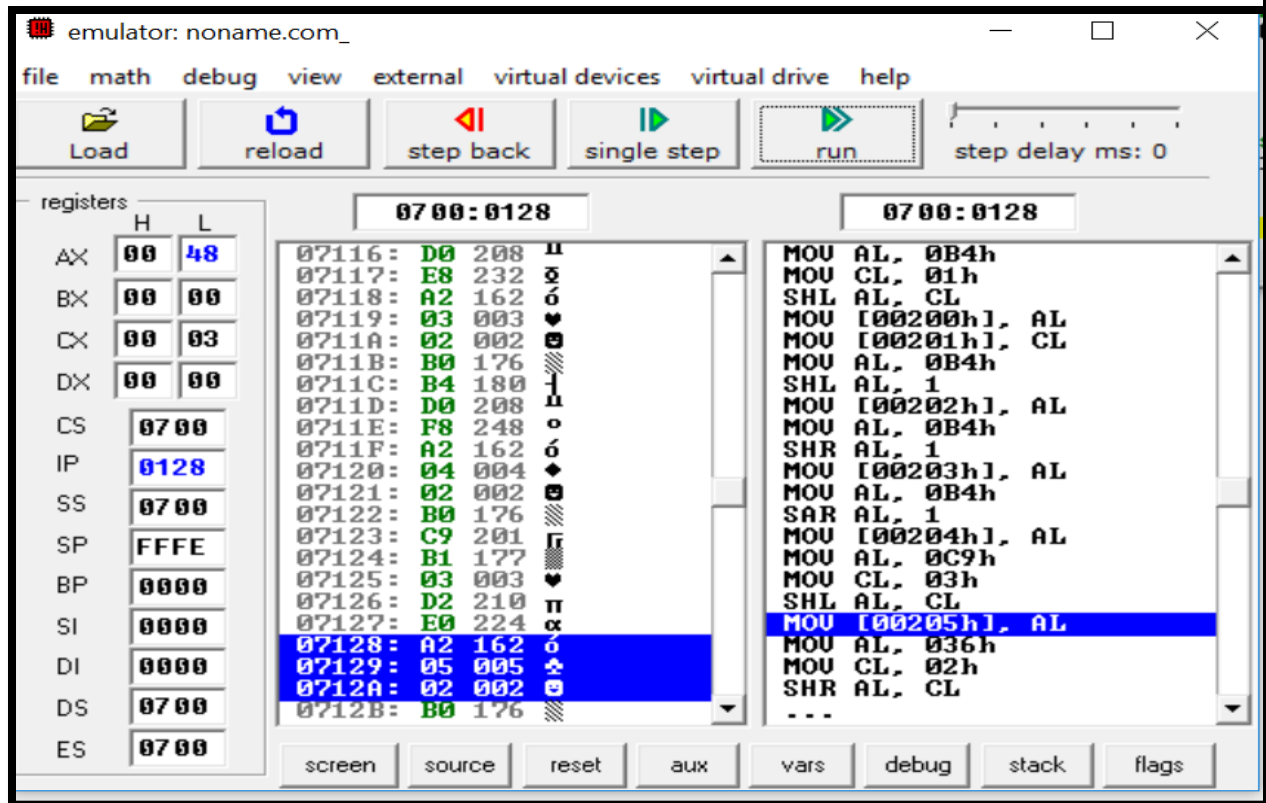
5. SAR AL,1

- **Expected Output:** AL=DAh, CF=0, OF=0.
 - *(Insert screenshot of Registers + Flags after SAR)*
- Screenshot 11: Registers/Flags after SAR**



6. Multi-bit Shift (SHL AL,3 or SHR AL,2)

- **Expected Output Example:** SHL by 3 → AL=48h.
 - *(Insert screenshot of Registers + Flags after multi-bit shift)*
- Screenshot 12: Registers/Flags after Multi-bit Shift**



7. Memory Dump Results

- **Expected Values:**
- 0200h → 68h (SHL)
- 0202h → 68h (SAL)
- 0203h → 5Ah (SHR)
- 0204h → DAh (SAR)
- 0205h → 48h (SHL by 3)
- 0206h → 0Dh (SHR by 2)

Screenshot 13: Memory Dump (0200h–0206h)

